TC-Lab: what remains in the error?

Conductance variation, radiation and discrete measurements

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A centered 10% conductance band improves the joint fit substantially, but nominal predictions for omitted protocols remain poor. Quantization-aware refitting retains input-linked and cross-channel dependence. The completed analyses therefore support conditional comparisons and identify limitations; they do not establish calibrated independent innovations.

Question	Finding	Status
Can modest run variation help?	A 10% conductance band gives T1/T2 RMSE 0.605/0.118 °C in the joint bin-likelihood fit.	Audited; little nominal protocol-transfer gain.
Does radiation explain the rest?	Free radiation gives 0.590/0.119 °C in the joint fit; bin nll improves by 3.810.	Profile checked; no calibrated radiation significance.
Are cold starts plausible?	Quantized hidden heater offsets remain below 0.515 °C, but exchange lag labels.	Startup sensitivity checked; equilibrium unverified.
Are measurements discrete?	All 21,614 temperatures match a shifted 0.3223 °C code grid.	Hardware evidence; threshold phase remains assumed.
Do lag labels identify physics?	At the loading boundary, a nominal forced-response lag swap changes T1 by at most 0.00430 °C.	Weak assignment; lowering the loading bound barely changes the mean.
Are residuals adequate for inference?	Deterministic code diagnostics retain input, phase and cross-channel structure.	Dependence persists after refitted reference checks.
Does the nominal fit transfer?	Omit one run: 1.750/0.754 °C; omit a protocol: 2.137/0.878 °C.	All twenty folds audited; local-normal propagation fails a shape check.

1 Records and model

We accept the same-device ENV2 provenance as the study assumption. The twelve distinct records comprise nine sine tests (three repeats at each of 30, 120 and 300 s) and three step tests. Duplicate aliases do not count as experiments. Only within-run time is available; Git history does not establish recovery time between runs.

Both temperatures enter a coupled heater/sensor model. With $i = 1, 2$ and the other heater denoted by k,

$$C_H \dot{H}_i = U_{a,j}(A_j - H_i) + U_s(S_i - H_i) + U_x(H_k - H_i) + \beta z_i, \tag{1}$$

$$C_S \dot{S}_i = U_s(H_i - S_i), \tag{2}$$

$$L\dot{z}_1 = u + c\,u(1 - u/100) - z_1, \qquad z_2 = 0. \tag{3}$$

The recorded command is held from its timestamp to the next update. The empirical curvature is not a measured electrical power law. Each run has $A_j = T_{1,j}(0) + a_j$, with $|a_j| \le 2\,°\mathrm{C}$. The baseline starts $H_i(0) = S_i(0) = T_{i,j}(0)$ and $z_1(0) = 0$.

Absolute power calibration is unavailable. Estimate scale-free quantities

$$K_0 = \beta/U_{a,0}, \quad t_A = C_H/U_{a,0}, \quad r_S = C_S/U_s, \quad r = U_s/U_{a,0}, \quad q = U_x/U_{a,0}.$$

A constant shared heater calibration multiplier is absorbed into gain. Temperature-only fitting does not identify absolute conductances or heat capacities without an independent scale.

One physical coefficient varies across runs:

$$U_{a,j} = U_{a,0}e^{\delta_j}, \quad \sum_j \delta_j = 0, \quad \log(1-b) \le \delta_j \le \log(1+b).$$

Thus $K_j = K_0 e^{-\delta_j}$, $t_{A,j} = t_A e^{-\delta_j}$, $r_j = re^{-\delta_j}$ and $q_j = qe^{-\delta_j}$, while r_S stays shared. The hard band preserves coupling between experiments. It is not a population distribution or a fitted random-effects variance.

2 What the matched fits show

The old short-lag/curved-command coupled fit has T1 RMSE 1.022 °C. Orthogonal error projections attribute 74.6% of its T1 squared error to run offsets and another 19.3% to run trends. The additional fundamental sine projection contributes only 0.64%. These are descriptive projections, not independent physical causes. The earlier description of a one-degree amplitude error confused total RMSE with sine amplitude mismatch.

Gaussian comparison	T1 RMSE, °C	T2 RMSE, °C
Shared short lag and curved command	1.022	0.231
Conductance band 5%	0.646	0.147
Conductance band 10%	0.639	0.115
Conductance band 20%	0.641	0.115
10% band, affine ambient	0.728	0.111
10% band, hidden heater starts	0.615	0.111
10% convective band, free radiation	0.633	0.115

Widening the band adds little beyond 10% in this comparison. Two 10% deviations reach their lower bound; protocol-aligned ratios are fitted behavior, not measured airflow. Likelihood and unweighted RMSE need not rank models alike. These fits reuse all twelve records.

The earlier T1-only positive three-pole benchmark, with command curvature and affine ambient, reached 0.374 °C RMSE. Its different output set and ambient formulation prevent a matched likelihood or AIC comparison with these joint fits. It remains historical prediction context, without a unique physical assignment of its three poles.

Radiation scale

Use Kelvin in $Q_{\mathrm{rad}} = \epsilon\sigma A_{\mathrm{rad}}(H^4 - T_r^4)$. For illustrative area 0.0012 m^2, surroundings at 23 °C and surface temperature 70.54 °C, emissivity scenarios 0.05–0.9 give 0.021–0.383 W. Against a linear radiative loss through ambient, matched at 50 °C, the nonlinear remainder is only 0.0021–0.0370 W. A locally linear radiative contribution can already reside in fitted U_a.

Metal alone does not determine emissivity. Finish, effective area, view factors and surface temperature are unknown; a sensor reading is not a known heater-surface temperature. Published TC-Lab models include convection, radiation and sensor dynamics [1]. Their values supply modeling context, not calibration for these records.

The free-radiation fit refits convection and gain to avoid double counting. Its normalized differential radiation coefficient at 50 °C is 0.0344 relative to nominal convection. Gaussian nll improves by 1.285. This is not a measured emissivity or a calibrated significance result.

Why assigning the two lags is difficult

At zero sensor loading, common/differential forced transfers factor as

$$G_{\pm}(s) = \frac{K}{(t_A s + 1 + \{0, 2q\})(1 + r_S s)(1 + Ls)}.$$

The output channels are $(G_+ + G_-)/2$ and $(G_+ - G_-)/2$. Exchanging r_S and L leaves these forced transfers unchanged in that limit.

At the fitted loading boundary $r = 0.005$, exchanging the nominal 14.98 s sensor lag and 2.69 s input lag changes the synthetic fast-sine T1/T2 waveforms by at most 0.00430/0.00185 °C. That calculation uses equal 20 °C starts and zero ambient offset. Unequal starts and ambient forcing can break the exchange. Repeated measurements can constrain sub-grid effects, so code spacing is not an absolute parameter-resolution limit. Lowering the positive loading bound from 0.005 to 0.0001 again reaches the bound and improves bin nll by only 0.01291; gain and fitted time scales barely change. This does not establish zero sensor heat capacity. The profiles below retain both lag regions.

3 Discrete observations and residual checks

The hardware data, rather than just simulator output, are quantized. The inferred code values are round$(0.3223n - 50, 2)$. Their grid does not identify pre-quantization noise or exact thresholds.

There is also a temperature-reference freedom in the linear model. When cold initial sensor states and ambient can shift together, changing nearest to floor bins translates every interval and predicted temperature by half a code, 0.16115 °C, without changing the likelihood. An independent synthetic check verifies this identity with slack ambient bounds. Fixed initial states, active ambient limits or absolute-temperature radiation can break it. A threshold phase comparison must distinguish this reference shift from dynamic information.

For the unreported analogue temperature,

$$Y_i^* = m_i(\psi) + X_i + v_i, \qquad X_i = e^{-\Delta t_i/\ell} X_{i-1} + w_i,$$

with stationary variance S, process variance $S(1 - e^{-2\Delta t_i/\ell})$ and independent $v_i \sim N(0, R)$. The likelihood integrates conditional observation density over each code bin. Forward quadrature and backward smoothing account for correlated latent errors. Multiplying marginal bin probabilities gives a different likelihood.

Holding the 10% mean fit fixed, bin-likelihood refitting changes analogue measurement SD from about 0.094 to 0.0255 °C for T1, and from 0.072 to 0.0325 °C for T2. T2 correlation length changes from 7.9 to 54.8 s. These are conditional noise estimates, not calibration measurements. The joint 10% bin-likelihood fit is now independently audited. Its T1/T2 RMSE is 0.605/0.118 °C, with $K_0 = 0.8732$ °C/% and $t_A = 174.59$ s. Gain is within 0.1% of the matched Gaussian estimate. The fitted noise is

Joint bin fit	T1	T2
Latent correlated SD, °C	0.5362	0.0681
Correlation length, s	658.28	62.56
Analogue measurement SD, °C	0.0258	0.0325

The joint 5% and 20% bands are also audited, with RMSE 0.637/0.153 °C and 0.606/0.118 °C. Their nll values are 6279.4250 and 6238.2167, versus 6238.2987 for 10%. Four 5% deviations reach bounds; the wider band is not selected from training likelihood alone. The joint radiation fit gives 0.590/0.119 °C and nll 6234.4890. Its improvement is larger than in the Gaussian approximation, but is neither a calibrated significance result nor evidence of measured emissivity.

Fisher's score identity supplies derivatives from smoothed latent moments [2]. Selected points require independent physical propagation, Gaussian bin rectangles, grid/support convergence and score stationarity. A repaired rare-bin integration failure remains a numerical failure; it is not evidence against the shared physical model.

Whitening checks combine independent randomized predictive residuals [3], serial and cross-channel dependence, input/phase dependence, and nonrandomized down/same/up code probabilities. Fifty fixed-model simulations are complete with no failed draw. Across twenty independent randomizations, median maximum serial correlation over lags 1–60 is 0.0248/0.0245 for T1/T2; maximum cross-channel correlation has median 0.0224. These are comparable to the fixed-model references. T1's nonrandomized code-transition discrepancy has reference upper-tail rank 0.0588. However, deterministic conditional expected Gaussian-PIT scores expose structure that the added PIT jitter diluted:

Statistic	Data	Reference 97.5%
T1 input/score correlation magnitude	0.04429	0.02261
T1 maximum phase projection	0.02983	0.01971
T2 maximum ACF, lags 1–60	0.04695	0.04039
Cross-channel maximum, lags ±10	0.04895	0.02540

Each descriptive upper-tail rank is 1/51. These scores are discrete and not unit-variance Gaussian residuals. The same calculation is applied to all fifty references. Removing run trends and three sine harmonics leaves cross correlation 0.04719, above its matched reference 97.5% value 0.02603. Projection has not resolved the dependence or identified its cause. Fixed-model references do not refit parameters or repeat selection, so these ranks are not calibrated composite-null p-values. Whitening remains unfinished.

Updating the residual state with earlier reported codes reduces whole-data one-step reported-temperature RMSE to 0.103/0.087 °C, compared with physical free simulation at 0.605/0.118 °C. At a horizon of at least 60 s, it is 0.255/0.110 °C. Parameters come from the same twelve records. These are feedback-assisted in-sample predictions; they do not validate the free dynamics or establish independent innovations.

Continuous-Gaussian joint sensitivities give process-driving correlation 0.0908 or measurement correlation −0.00692 when estimated separately. Estimating both gives 0.11587/−0.02463 and improves Gaussian nll by 6.14891. All have independent dense-covariance audits. This is a comparison on discrete temperatures using a continuous likelihood; it does not establish calibrated joint-bin probabilities or an environmental mechanism.

4 Uncertainty and identifiability

The independently audited information matrix distinguishes the centered tangent space from the selected active constraint face. Conditional local standard errors are

Coordinate	IID analogue	Correlated analogue	Joint bin
$\log K$	0.0007263	0.0033010	0.0033620
$\log t_A$	0.0005152	0.0030056	0.0030888
$\log L$	0.0479381	0.0309354	0.0324303

The first two columns hold covariance known and use identical analogue marginal variance. Correlation increases gain and thermal-time SEs by 4.55 and 5.83, while reducing the short-lag SE to 0.645 of its IID value. The bin column also estimates noise nuisances; its difference does not isolate quantization. These are selected-basin, active-face approximations, not calibrated intervals.

T1 log latent SD and log correlation length have local correlation 0.9845. Conditioning the residual process on the initial code changes their estimates from 0.5362 °C/658.3 s to 0.8763 °C/1761.9 s, while $2S/\ell$ changes only 0.22% and the mean parameters barely change. The data constrain the diffusion rate more clearly than its stationary variance/length split.

Allowing each cold heater/sensor pair within its first reported bin adds 24 bounded coordinates. Its audited nll falls to 6209.5536, gain to 0.8697 °C/% and short lag to 2.72054 s; several initial states reach edges. T1 free-simulation RMSE increases to 0.655 °C, while T2 falls to 0.113 °C. Better correlated-bin likelihood does not imply better noise-free simulation. These bin bounds do not integrate an initial measurement-noise posterior.

An observation-space check prevents a bin-reference shift from masquerading as improved prediction. The freely initialized nearest/floor fits differ in expected reported output by less than 0.00030 °C. Exact half-grid translation preserves those expectations within 3.1×10^{-12} °C. Their different raw code-minus-analogue-reference RMSE therefore does not demonstrate different predictive quality. The nearest free-initial fit's reported-code RMSE is 0.655/0.108 °C; stationary analogue noise and code integration are included.

Hidden-heater-only ±2 and ±5 °C sensitivities coincide, with maximum selected offset 0.514 °C. They favor sensor time 2.667 s and input lag 14.898 s, reversing the baseline labels. Their reported-code RMSE is 0.682/0.109 °C, against baseline 0.605/0.114 °C. A better likelihood did not improve T1 free prediction. The representative startup profile below retains this assignment near zero. These changes are distinct from the earlier Gaussian hidden-start comparison.

Profiles refit all ambient, conductance and noise nuisances. Sensor-loading bounds, initial-state sensitivities and lag alternatives make routine Wald intervals unsafe without those checks.

The nine-point lag grid passed independent audits. Twice the nll increase is 552.445 at $L = 7$ s but falls to 6.253 at 15 s, where the fitted sensor time is 2.643 s. A further audited fixed point at $L = 14.79006$ s fits sensor time 2.7314 s and gives twice the nll increase 4.78730. Both starts converge, without improving the reference. These checks expose a second region; they do not locate its free minimum, interpolate endpoints or calibrate an exclusion of that lag assignment.

What the nuisance profiles resolve

All 28 loading, run-conductance, disturbance-length and radiation grid points passed independent numerical audits. Raising loading $U_s/U_{a,0}$ from 0.005 to 0.1 or 0.3 increases twice nll by only 0.577 or 2.316. For step run 1, raising its conductance ratio from the lower bound 0.9 to 0.925 costs 17.215. The fitted band therefore constrains a consequential run effect; it has not estimated a distribution of airflow.

Radiation's optimum normalized coefficient is 0.11530. Setting it to zero costs twice nll 7.619 with all permitted nuisances refitted. This supports a conditional model improvement, without a calibrated boundary test or an emissivity estimate. The largest coefficient tested, 4, reaches gain and noise-length bounds. The coarse grid and two starts do not prove global minima.

The startup grid fixes only the first 300 s sine run's H1 offset and refits all other 23 hidden starts, dynamics, ambient, conductance and noise terms. Five of six points have accepted, independently audited fits. At zero offset, twice nll rises 6.447; sensor/input times remain about 2.678/14.881 s. The $-2\,^{\circ}\mathrm{C}$ task exhausted both declared starts and has no accepted profile value. No accepted point beats the startup reference. This is a sensitivity of one adaptively selected coordinate, rather than a confidence interval or an all-cold-start comparison.

Fifty refitted recovery collections

All fifty declared generated collections and all one hundred free/true-gain point audits passed, with no missing outcome. The nominal 95% gain likelihood-ratio region includes the generating value in 49/50 collections. Its independently inverted exact-binomial 95% Monte Carlo interval is 89.35–99.95%. This is conditional on the generating model, initialization, bins, bounds and fitting algorithm. It does not establish real-device coverage.

Fifteen refits exchange the sensor/input lag labels. Unordered fast and slow 2.5–97.5% repeated-fit ranges are 2.547–2.898 s and 14.419–15.576 s. These are estimator quantiles, not physical confidence intervals. Gain's sampling SD is 0.898 times its local SE; log thermal time spreads 4.780 times farther than the selected-face approximation. Named lag-coordinate spreads mix the two label regions and should not be summarized as one Gaussian error.

Refitting preserves the residual inadequacy finding. The reference 97.5% values for T1 input, T1 phase, T2 ACF and cross-channel diagnostics become 0.01689, 0.02146, 0.03894 and 0.03014, respectively. The hardware values in the earlier table exceed every one of the fifty refitted references. These are descriptive selected-diagnostic ranks, without multiple-testing or structure-selection calibration. Randomized PIT agreement has not established approximately IID innovations.

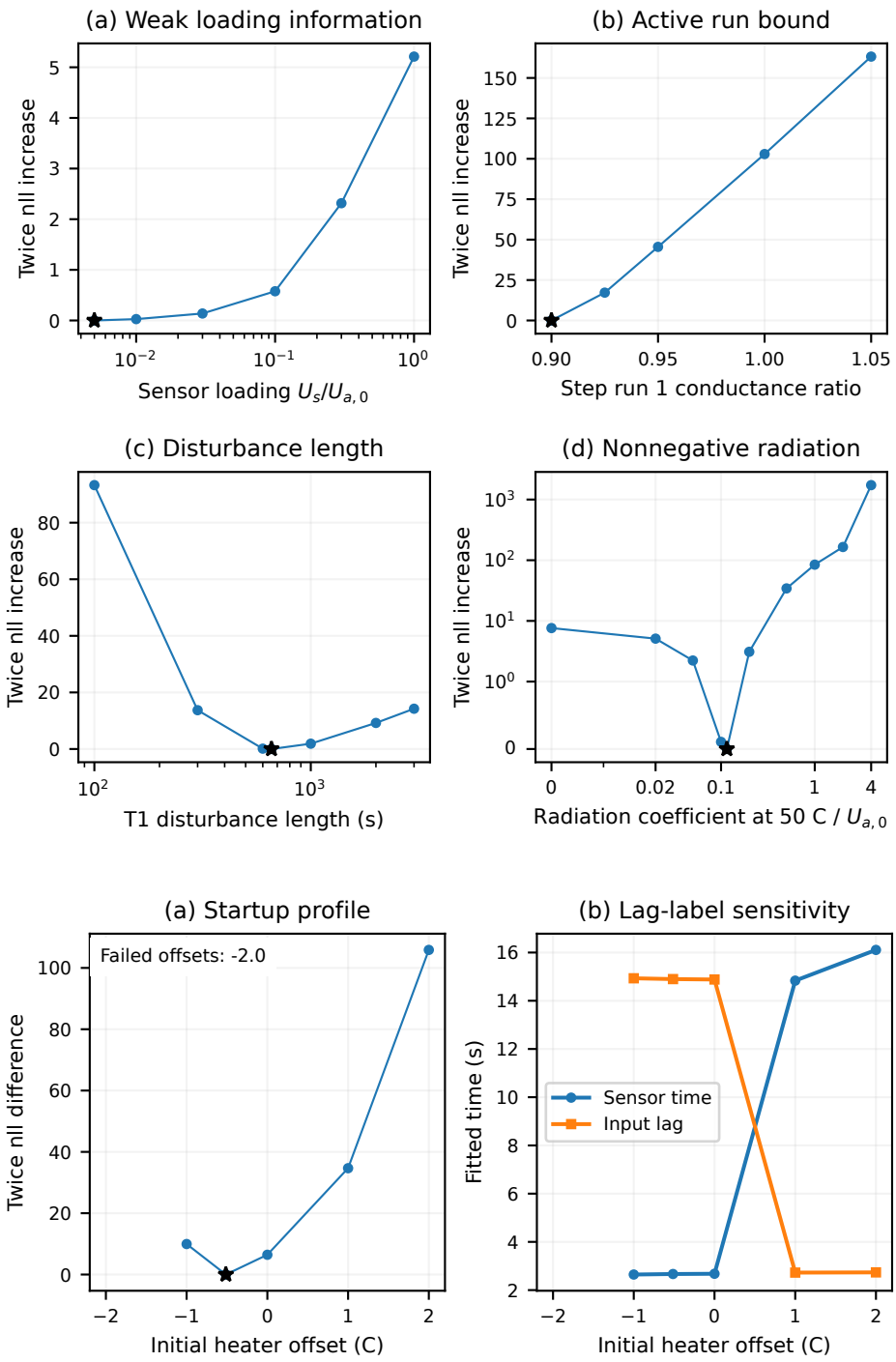


Figure 1: Audited nuisance profiles. The upper panels show all 28 grid points; radiation axes use symlog scales. The lower panels fix one hidden-heater initial offset. The failed $-2\,^{\circ}\mathrm{C}$ task has no plotted deviance. All other permitted coordinates are refitted. No routine chi-square cutoff is assigned.

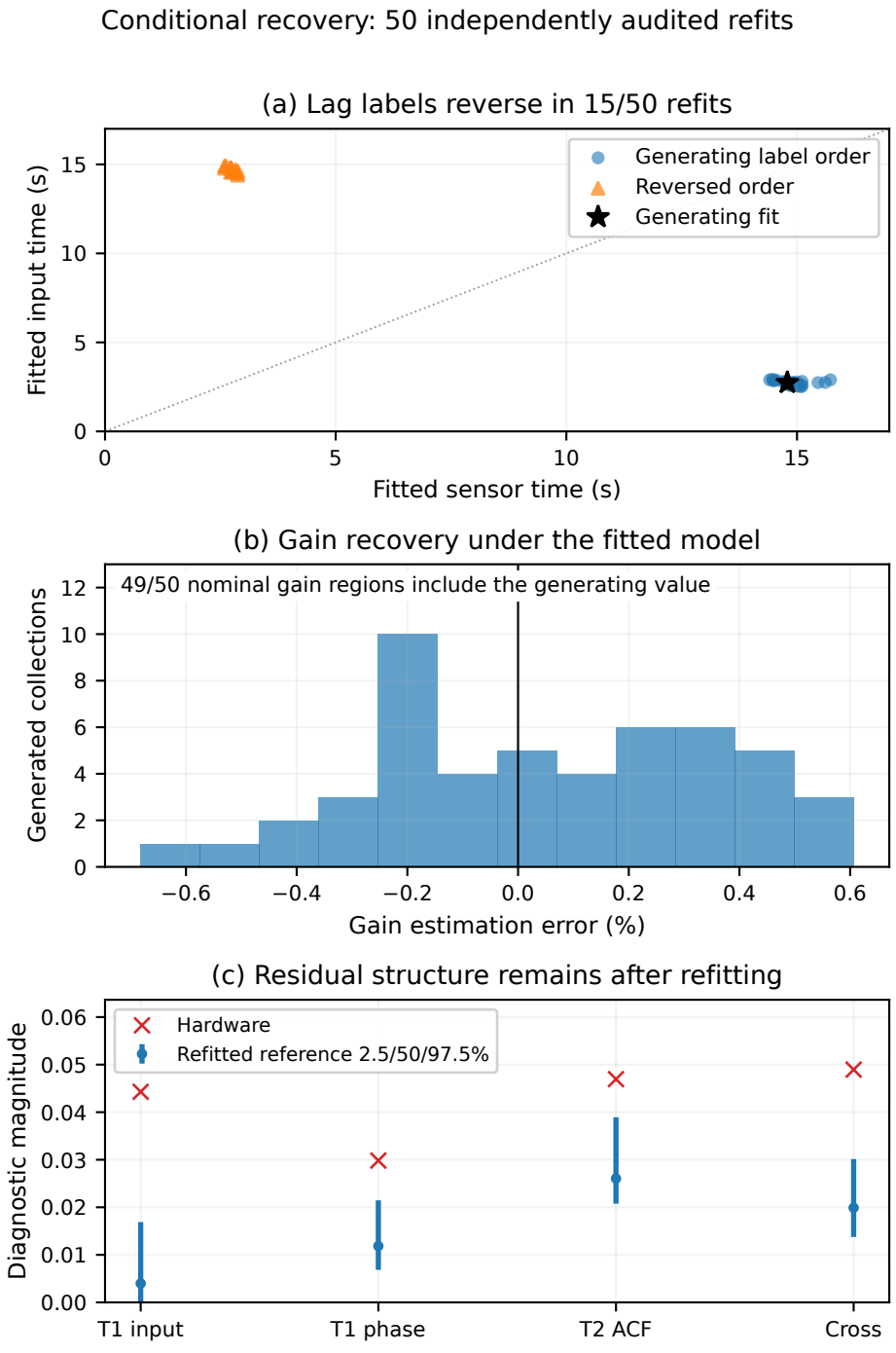


Figure 2: Conditional recovery from fifty fully audited refits. Lag labels exchange while the two time scales remain separated. The gain calculation checks nominal scalar LR coverage only. Hardware residual diagnostics exceed the refitted ranges; the analysis has not resolved their cause.

Retrospective transfer across all twenty folds

Twelve fits omit one run each. Four band-model fits and four shared-conductance fits omit one protocol each (30, 120 or 300 s sine, or step). All twenty selected fits, information matrices, noise and parameter forecasts passed independent numerical audits. No later test temperatures estimate ambient, conductance or physical parameters; the first temperature pair initializes the physical states. Each record appears once per comparison group, 10,795 scored readings per channel. All records informed structure selection and initial guesses, so these comparisons are retrospective.

Nominal reported-mean RMSE, °C	T1	T2
Band: omit one run	1.750	0.754
Band: omit protocol	2.137	0.878
Shared: omit protocol	2.178	0.896

Conductance variation improves the all-data fit much more than nominal protocol predictions. Run mean errors account for 84.9/72.5% of leave-run T1/T2 SSE; removing them leaves RMSE 0.680/0.395 °C. Omitted protocols give 85.7/73.3% and 0.809/0.454 °C. Offset removal uses test outcomes and cannot serve as a blind correction. Shared-conductance omitted-protocol RMSE is only slightly worse than the band model's.

Range and group	Coverage, % T1/T2	Mean width, °C T1/T2
Band/run, noise	46.01/47.56	2.093/0.291
Band/run, conditions + noise	100.00/99.94	9.160/5.170
Band/run, parameters	100.00/99.94	11.120/6.048
Band/protocol, noise	31.84/36.48	2.006/0.302
Band/protocol, conditions + noise	99.57/99.95	9.068/5.174
Band/protocol, parameters	99.64/99.95	9.274/5.346
Band/protocol, reference stress	100.00/99.95	10.380/5.737
Shared/protocol, noise	49.87/53.11	3.610/0.812
Shared/protocol, conditions + noise	94.01/99.81	6.771/3.968
Shared/protocol, parameters	94.65/100.00	7.115/4.286

Nominal conditions use conductance ratio one and ambient offset zero. Condition ranges allow ±2 °C ambient and, for the band model, ±10% conductance. They combine hard scenarios with marginal noise quantiles. The table's "parameters" rows also mix 256 box-truncated local normal draws of physical and noise nuisance coordinates. They fix active bounds and one likelihood basin. Their numerical arithmetic passed; their sampling calibration has not been established. Mixture audits check all components' quantile brackets at declared times and selected trajectories, without independently recomputing every component history. Broad hard ranges and high pointwise coverage do not define a probability distribution over run conditions. Correlated readings are not independent validation replicates.

The complete local-parameter calculations change omitted-protocol band-model mean widths from 9.068/5.174 °C to 9.274/5.346 °C. Leave-run and shared-model changes are larger. Shared-prefix 64/256 comparisons also retain appreciable endpoint changes:

64/256 comparison	Max change, °C T1/T2	Endpoints changed, % T1/T2
Band/run	1.61/0.65	19.11/11.08
Band/protocol	0.65/0.33	8.71/5.39
Shared/protocol	1.62/0.65	16.54/13.74

The first prototype, omitting the three fast-sine runs, changes widths from 9.119/5.134 °C to 9.223/5.206 °C. Its 64/256 comparison changes 2.26/2.69% of endpoints by at most one code (0.33 C), and its local log-input-lag SE is 0.1232. This single-fold result does not describe all twenty folds. The complete table above reports the larger sensitivities elsewhere.

Why the local mixtures remain exploratory

Positive numerical curvature alone did not validate a local normal law. A declared check evaluated 16 saved draws in each of three cases: the full fit, omission of 30 s sines, and the run-omission fit with largest conditional log-thermal-time SE, selected using training geometry. At that last fit, a representative draw predicts twice nll increase 34.276 from the local Taylor expansion; independent evaluation gives 201.731. This discrepancy survives 128/256/512/1024-node and tail-support checks, with confirmed absolute error 167.454. The local approximation fails along a curved weak nuisance direction, even with a numerically audited Hessian.

The most extreme saved draw fails the declared integration-convergence gate even at 1024 nodes and wider support. Its precise deviance is withheld. No draws were clipped, discarded or reweighted to make the ranges agree. Increasing Monte Carlo count would not repair the demonstrated approximation error. The parameter-inclusive rows therefore quantify sensitivity to an unvalidated local law; they are not calibrated 95% prediction intervals. The fifty-refit gain check supports only its specified scalar/model case.

Each training subset defines a different geometric nominal conductance. Compare invariant combinations $K_0/t_A = \beta/C_H$, $r/t_A = U_s/C_H$ and $q/t_A = U_x/C_H$ before treating raw gain shifts as physical change. Centering a hypothetical twelve-run cohort permits an omitted run's ratio to a nine-run training nominal in $[0.86894, 1.13551]$. The four audited reference-stress forecasts propagate that range without changing the training fits: mean widths become 10.380/5.737 °C. This is a scenario stress check, without a transported-cohort likelihood or corrected coverage calibration.

5 Questions for scientific review

One bounded conductance effect captures much of the between-run mismatch. It does not identify airflow as its cause, and its benefit transfers weakly across protocols. Radiation adds modest conditional improvement. Small hidden initial offsets change physical lag labels, while the two fitted time scales remain near 3 and 15 s. Discrete measurement modeling changes noise estimates and exposes structure that randomized residual jitter can conceal.

The residual model has not achieved approximately calibrated independent innovations. The hardware input/phase and cross-channel failures, the lag exchange and the local-normal shape failure limit uncertainty claims. A justified next analysis would compare a quantized multichannel disturbance model and a less symmetric physical model against the same diagnostics, without treating a higher likelihood as adequate whitening. Neither a unique cause nor a physical emissivity or electrical power law has been identified.

A later hood experiment should log ambient temperature, absolute time, previous run recovery, airflow/sash settings, firmware, board/sensor identity, heater P settings and electrical power. A hood can add forced or fluctuating convection; it does not automatically supply a constant environment.

Evidence and scope

The committed source, raw-data snapshots, multistart outcomes, numerical failures, independent checks and complete fold index are retained in the remaining-error study. The Markdown companion links the detailed methods, profiles, recovery and transfer evidence. Figures and report values are checked separately from scientific interpretation. These scoped analyses stop for Alex's review; no hardware, DoE, pair enumeration or new email is part of this study.

References

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